

Concavity, Curvature, and the Second Derivative

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Outline (75 minutes)

1. **The Second Derivative** – Definition and computation (10 min)
2. **Concavity** – Concave up, concave down, and their meaning (20 min)
3. **Inflection Points** – Where concavity changes (15 min)
4. **The Second Derivative Test** – Classifying extrema (15 min)
5. **Curvature** – The geometry of bending (15 min)

1 The Second Derivative

1.1 Definition

Definition 1.1. The **second derivative** of a function $f(x)$ is the derivative of its first derivative:

$$f''(x) = \frac{d}{dx} (f'(x)) = \frac{d^2 f}{dx^2}$$

Notation: We use $f''(x)$, $\frac{d^2 f}{dx^2}$, or $D_x^2 f(x)$ interchangeably.

1.2 Computing Second Derivatives

Example 1: $f(x) = x^3 - 6x^2 + 9x + 1$

$$f'(x) = 3x^2 - 12x + 9$$

$$f''(x) = 6x - 12$$

Example 2: $f(x) = e^{-x^2}$ (Gaussian)

$$f'(x) = -2xe^{-x^2}$$

$$f''(x) = -2e^{-x^2} + (-2x)(-2xe^{-x^2})$$

$$= -2e^{-x^2} + 4x^2e^{-x^2}$$

$$= (4x^2 - 2)e^{-x^2}$$

Example 3: $f(x) = \sin(x)$

$$f'(x) = \cos(x)$$

$$f''(x) = -\sin(x)$$

1.3 What Does the Second Derivative Measure?

First derivative: Rate of change of the function itself (slope)

Second derivative: Rate of change of the slope

- If $f''(x) > 0$: The slope is *increasing*
- If $f''(x) < 0$: The slope is *decreasing*
- If $f''(x) = 0$: The slope is (momentarily) constant

2 Concavity

2.1 Definitions

Definition 2.1. A function f is **concave up** on an interval if $f''(x) > 0$ for all x in that interval.

A function f is **concave down** on an interval if $f''(x) < 0$ for all x in that interval.

Visual interpretation:

- **Concave up:** The graph curves upward like a cup ($\cup$). “Holds water.”
- **Concave down:** The graph curves downward like a cap ($\cap$). “Spills water.”

Tangent line interpretation:

- Concave up: The graph lies *above* its tangent lines
- Concave down: The graph lies *below* its tangent lines

2.2 Example: Analyzing $f(x) = x^3$

Compute derivatives:

$$\begin{aligned} f(x) &= x^3 \\ f'(x) &= 3x^2 \\ f''(x) &= 6x \end{aligned}$$

Analyze concavity:

- For $x > 0$: $f''(x) = 6x > 0 \Rightarrow$ concave up
- For $x < 0$: $f''(x) = 6x < 0 \Rightarrow$ concave down
- At $x = 0$: $f''(0) = 0 \Rightarrow$ concavity changes!

2.3 Example: The Parabola $f(x) = x^2$

Compute:

$$\begin{aligned}f'(x) &= 2x \\f''(x) &= 2\end{aligned}$$

Conclusion: $f''(x) = 2 > 0$ for all x , so $f(x) = x^2$ is **always concave up**.

This confirms the intuition that a parabola opening upward is “cup-shaped” everywhere.

2.4 Example: The Gaussian $f(x) = e^{-x^2}$

From earlier: $f''(x) = (4x^2 - 2)e^{-x^2}$

Since $e^{-x^2} > 0$ always, the sign of $f''(x)$ depends on $4x^2 - 2$:

$$\begin{aligned}4x^2 - 2 > 0 &\Leftrightarrow x^2 > \frac{1}{2} \Leftrightarrow |x| > \frac{1}{\sqrt{2}} \\4x^2 - 2 < 0 &\Leftrightarrow |x| < \frac{1}{\sqrt{2}}\end{aligned}$$

Conclusion:

- $|x| < 1/\sqrt{2} \approx 0.707$: concave down (near the peak)
- $|x| > 1/\sqrt{2}$: concave up (in the tails)

3 Inflection Points

3.1 Definition

Definition 3.1. An **inflection point** is a point where the concavity changes from up to down or from down to up.

Necessary condition: If $x = c$ is an inflection point, then $f''(c) = 0$ or $f''(c)$ is undefined.

Warning: $f''(c) = 0$ is *necessary but not sufficient*. We must verify that concavity actually changes.

3.2 Finding Inflection Points: Procedure

1. Find $f''(x)$
2. Solve $f''(x) = 0$ for candidate points
3. Check that $f''(x)$ changes sign at each candidate
4. If sign changes, it's an inflection point

3.3 Example: $f(x) = x^3 - 3x^2 + 2$

Step 1: Compute derivatives.

$$\begin{aligned}f'(x) &= 3x^2 - 6x \\f''(x) &= 6x - 6 = 6(x - 1)\end{aligned}$$

Step 2: Solve $f''(x) = 0$.

$$6(x - 1) = 0 \Rightarrow x = 1$$

Step 3: Check sign change.

- For $x < 1$: $f''(x) = 6(x - 1) < 0$ (concave down)
- For $x > 1$: $f''(x) = 6(x - 1) > 0$ (concave up)

Step 4: Conclusion.

Yes, concavity changes at $x = 1$. The inflection point is $(1, f(1)) = (1, 0)$.

3.4 Example: $f(x) = x^4$ (No Inflection Point)

Compute:

$$\begin{aligned}f'(x) &= 4x^3 \\f''(x) &= 12x^2\end{aligned}$$

Solve $f''(x) = 0$: $x = 0$

Check sign:

- For $x < 0$: $f''(x) = 12x^2 > 0$
- For $x > 0$: $f''(x) = 12x^2 > 0$

Conclusion: $f''(x) \geq 0$ everywhere—no sign change! The point $x = 0$ is *not* an inflection point; the function is concave up everywhere.

4 The Second Derivative Test

4.1 Finding Local Extrema

Recall: Local maxima and minima occur where $f'(x) = 0$ (critical points).

Question: How do we determine if a critical point is a max, min, or neither?

Theorem 4.1 (Second Derivative Test). Suppose $f'(c) = 0$ (i.e., c is a critical point).

- If $f''(c) > 0$, then f has a **local minimum** at $x = c$
- If $f''(c) < 0$, then f has a **local maximum** at $x = c$
- If $f''(c) = 0$, the test is **inconclusive**

Intuition:

- $f''(c) > 0$ means concave up at c — the critical point is a valley (minimum)
- $f''(c) < 0$ means concave down at c — the critical point is a peak (maximum)

4.2 Example: $f(x) = x^3 - 12x + 5$

Step 1: Find critical points.

$$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x - 2)(x + 2) = 0$$

Critical points: $x = -2$ and $x = 2$

Step 2: Compute second derivative.

$$f''(x) = 6x$$

Step 3: Apply second derivative test.

- At $x = -2$: $f''(-2) = -12 < 0 \Rightarrow$ **local maximum**
- At $x = 2$: $f''(2) = 12 > 0 \Rightarrow$ **local minimum**

Step 4: Find the extrema values.

$$f(-2) = (-2)^3 - 12(-2) + 5 = -8 + 24 + 5 = 21 \quad (\text{local max})$$

$$f(2) = 8 - 24 + 5 = -11 \quad (\text{local min})$$

4.3 Example: When the Test is Inconclusive

Consider $f(x) = x^4$:

$$f'(x) = 4x^3 = 0 \Rightarrow x = 0$$

$$f''(x) = 12x^2, \quad f''(0) = 0 \quad (\text{inconclusive})$$

Resolution: Use the first derivative test instead.

- For $x < 0$: $f'(x) = 4x^3 < 0$ (decreasing)
- For $x > 0$: $f'(x) = 4x^3 > 0$ (increasing)

Since f decreases then increases, $x = 0$ is a local minimum.

5 Curvature

5.1 Beyond Concavity: Measuring “Bendiness”

Observation: The second derivative tells us the *direction* of bending (up or down) but not how *sharply* a curve bends.

Definition 5.1. The **curvature** κ at a point measures how quickly the curve changes direction. It is the reciprocal of the radius of the **osculating circle** (the circle that best fits the curve at that point).

Formula:

$$\kappa(x) = \frac{|f''(x)|}{(1 + [f'(x)]^2)^{3/2}}$$

Key properties:

- Large κ = tight curve (small radius)
- Small κ = gentle curve (large radius)
- $\kappa = 0$ = straight line

5.2 Curvature of a Circle

For a circle of radius R : The curvature is constant:

$$\kappa = \frac{1}{R}$$

Check: A small circle (R small) has large curvature; a large circle (R large) has small curvature. A straight line is a “circle of infinite radius,” so $\kappa = 0$.

5.3 Example: Curvature of $f(x) = x^2$

Compute derivatives:

$$\begin{aligned} f'(x) &= 2x \\ f''(x) &= 2 \end{aligned}$$

Apply curvature formula:

$$\kappa(x) = \frac{|2|}{(1 + (2x)^2)^{3/2}} = \frac{2}{(1 + 4x^2)^{3/2}}$$

At specific points:

x	$\kappa(x)$	Interpretation
0	$\frac{2}{1} = 2$	Maximum curvature at vertex
1	$\frac{2}{5^{3/2}} \approx 0.179$	Less curved
2	$\frac{2}{17^{3/2}} \approx 0.029$	Nearly straight

Key insight: Even though $f''(x) = 2$ is constant (uniform concavity), the *curvature* varies—it's greatest at the vertex and decreases as we move away.

5.4 Application: Highway Design

Problem: When designing roads, curves must be introduced gradually to prevent sudden steering adjustments.

Key principle: The curvature should change continuously—ideally, $d\kappa/ds$ (rate of change of curvature with respect to arc length) should be small.

Clothoid spirals: Highway engineers use curves where curvature increases linearly with distance. These are called *Euler spirals* or *clothoids*.

Why it matters:

- Sudden curvature changes cause jerky steering
- Centrifugal force spikes can cause accidents
- Smooth curvature transitions improve safety and comfort

5.5 Radius of Curvature

Definition 5.2. The **radius of curvature** at a point is:

$$R(x) = \frac{1}{\kappa(x)} = \frac{(1 + [f'(x)]^2)^{3/2}}{|f''(x)|}$$

Interpretation: $R(x)$ is the radius of the circle that best approximates the curve at point x .

Example: For $f(x) = x^2$ at $x = 0$:

$$R(0) = \frac{1}{\kappa(0)} = \frac{1}{2}$$

The parabola at its vertex is approximated by a circle of radius $1/2$.

6 Summary

6.1 Key Concepts

1. **Second derivative** $f''(x)$: Rate of change of the slope
2. **Concavity:**
 - $f''(x) > 0$: concave up (cup)
 - $f''(x) < 0$: concave down (cap)
3. **Inflection point:** Where concavity changes (requires $f''(x) = 0$ and sign change)
4. **Second Derivative Test:** At critical point c where $f'(c) = 0$:
 - $f''(c) > 0 \Rightarrow$ local minimum
 - $f''(c) < 0 \Rightarrow$ local maximum
5. **Curvature:** $\kappa = \frac{|f''(x)|}{(1 + [f'(x)]^2)^{3/2}}$

6.2 Practice Exercises

Exercise 1: For $f(x) = x^4 - 4x^3 + 6x^2$:

- (a) Find all inflection points.
- (b) Determine intervals of concavity.

Exercise 2: Use the second derivative test to classify all critical points of $g(x) = x^3 - 6x^2 + 9x + 2$.

Exercise 3: Find the curvature of $f(x) = \sin(x)$ at $x = 0$ and $x = \pi/2$.

Exercise 4: A road follows the path $y = \ln(x)$ for $x > 0$. Find the point of maximum curvature.

Exercise 5: Prove that for the exponential function $f(x) = e^x$, the curvature is maximized at $x = -\frac{1}{2} \ln(2)$.

Worked Solutions

Exercise 1 (Solution): $f(x) = x^4 - 4x^3 + 6x^2$. Then $f'(x) = 4x^3 - 12x^2 + 12x = 4x(x^2 - 3x + 3)$ and $f''(x) = 12x^2 - 24x + 12 = 12(x^2 - 2x + 1) = 12(x - 1)^2$.

(a) Inflection points: Solve $f''(x) = 0 \Rightarrow 12(x - 1)^2 = 0 \Rightarrow x = 1$. Check sign change: for $x < 1$, $f''(x) = 12(x - 1)^2 > 0$; for $x > 1$, $f''(x) > 0$. So f'' does not change sign—there is **no inflection point** (the concavity is the same on both sides of $x = 1$; f is concave up everywhere).

(b) Intervals of concavity: $f''(x) = 12(x - 1)^2 \geq 0$ for all x . So f is **concave up on** $(-\infty, \infty)$. $\square$

Exercise 2 (Solution): $g(x) = x^3 - 6x^2 + 9x + 2$. Then $g'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3) = 0$ at $x = 1$ and $x = 3$. So critical points: $x = 1, x = 3$. Second derivative: $g''(x) = 6x - 12$. So $g''(1) = -6 < 0 \Rightarrow$ **local maximum** at $x = 1$; $g''(3) = 6 > 0 \Rightarrow$ **local minimum** at $x = 3$. $\square$

Exercise 3 (Solution): For $f(x) = \sin(x)$, we have $f'(x) = \cos(x)$ and $f''(x) = -\sin(x)$. Curvature: $\kappa(x) = \frac{|f''(x)|}{(1 + [f'(x)]^2)^{3/2}}$.

At $x = 0$: $f'(0) = 1, f''(0) = 0$, so $\kappa(0) = \frac{0}{(1+1)^{3/2}} = 0$.

At $x = \pi/2$: $f'(\pi/2) = 0, f''(\pi/2) = -1$, so $\kappa(\pi/2) = \frac{1}{(1+0)^{3/2}} = 1$. $\square$

Exercise 4 (Solution): For $f(x) = \ln(x)$ ($x > 0$), $f'(x) = \frac{1}{x}$ and $f''(x) = -\frac{1}{x^2}$. So $\kappa(x) = \frac{1/x^2}{(1+1/x^2)^{3/2}} = \frac{1}{x^2} \cdot \frac{x^3}{(x^2+1)^{3/2}} = \frac{x}{(x^2+1)^{3/2}}$. To maximize κ , set $\frac{d\kappa}{dx} = 0$. One finds that the maximum occurs when $x = \frac{1}{\sqrt{2}}$. So the point of maximum curvature is $(\frac{1}{\sqrt{2}}, \ln(\frac{1}{\sqrt{2}}))$. $\square$

Exercise 5 (Solution): For $f(x) = e^x$, $f'(x) = e^x$ and $f''(x) = e^x$. So $\kappa(x) = \frac{e^x}{(1+e^{2x})^{3/2}}$. To find where κ is maximized, set $\frac{d\kappa}{dx} = 0$. After simplifying, this gives $1 - 2e^{2x} = 0$, i.e., $e^{2x} = \frac{1}{2}$, so $2x = \ln(1/2) = -\ln 2$, hence $x = -\frac{1}{2} \ln(2)$. One can verify that $\kappa'' < 0$ at this point, so the curvature is maximized at $x = -\frac{1}{2} \ln(2)$. $\square$